FIRST REAL OUTDOOR TEST — OFF THE GRID

From the car boot to live data in 8 minutes

On Sunday 8 November 2026 the MODE-S Wind Tactical station was taken off its balcony, driven to a forest clearing, set up on its tripod and powered from a portable battery — no mains, no fixed network. It joined the MODE-S Wind server over 4G and fed live replies as **MODES-RX-02**, was packed back into the car, and in the afternoon went back to the balcony and on line again by itself.

8 min
arrival photo → live data on the server

≈ 5 min
power off → packed, ready to drive

10 W
the whole station from a 230 V outlet

13.8 h
left on the battery at 71 % (its display)

What was tested

The question was simple: *is the station really as quick to put up and take down as it is meant to be, away from home, with nothing but what fits in a car?* Everything was carried in the boot of an ordinary family car together with the weekend's shopping: the tripod, the mast with the 1090 MHz antenna, the enclosure with the receiver, the Raspberry Pi 5 and the 4G router, the coax and a power cable — and an **EcoFlow River 3 Plus** portable power station (286 Wh $LiFePO_4$ battery, 230 V AC outlet) standing in the open boot. On site: stand the tripod up, raise the mast, connect the coax, run the power cable from the boot, switch on. Nothing was typed in or configured.


12:38:58 — DEPLOYED

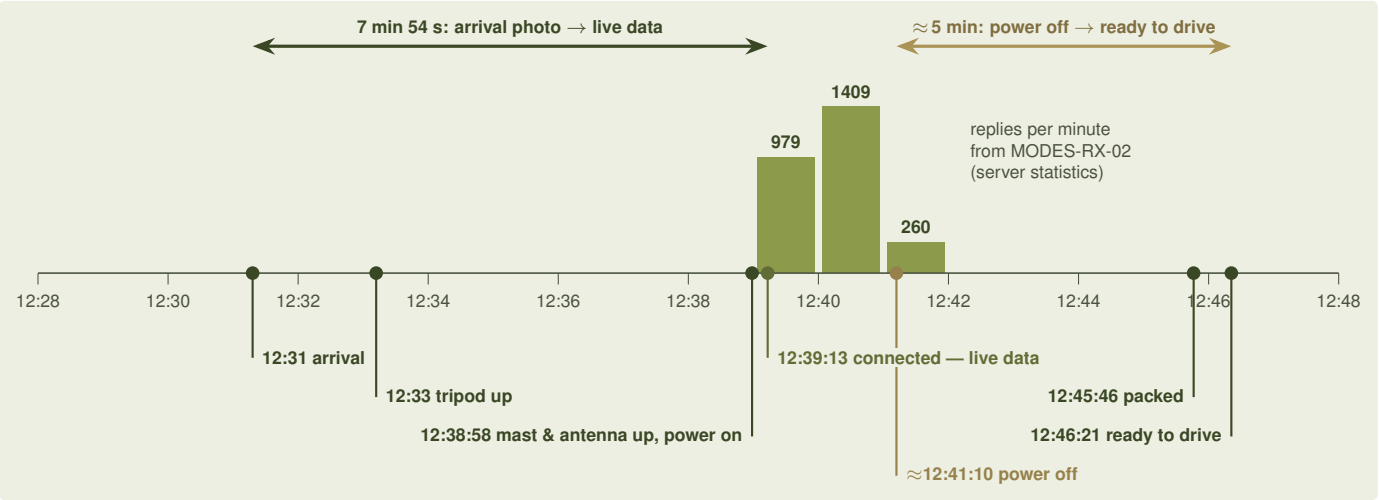

12:39:11 — POWER FROM THE CAR BOOT

Result in short

- **Up and feeding in minutes.** The first photo at the site was taken at 12:31; at 12:39:13 the server logged *MODES-RX-02: connected*, and 26 seconds later its health check showed the feed live at 24 messages a second. The station found the 4G network, opened its tunnel to the server and started feeding entirely by itself.
- **Real data, straight into the merged picture.** In about two minutes on the air it delivered 2 648 replies from 9 aircraft — ADS-B positions and velocities, and Comm-B weather registers (BDS 4,0 / 5,0 / 6,0) from which MODE-S Wind derives wind and temperature. The server's health check: *22.2 % of its replies heard only here* — by no other receiver.
- **Down in minutes.** Power off at about 12:41; at 12:45:46 everything was back in the boot and at 12:46 the car was ready to drive.
- **Off-grid power is no problem.** The whole station drew **9–10 W** from the battery's 230 V outlet. The battery's own estimate at 71 % charge: 13.8 hours left — a full day of feeding from one charge.
- **Back home, back on line by itself:** on the balcony again, it reconnected at 15:03:54 and fed 68 messages a second.

Chronology — 8 November 2026 (local time, UTC + 2)

The times come from the photos (the phone's clock) and from the MODE-S Wind server's log and statistics (UTC, given in local time here).



Local	UTC	Event	Source
11:59:42	09:59:42	Station switched off on the balcony for transport (MODES-RX-02 connection lost)	server log
12:31:19	10:31	Arrival at the test site, the station packed in the boot	photo
12:33:13	10:33	Tripod standing with the enclosure on it	photo
12:38:58	10:39	Mast and antenna up, coax connected, power cable from the battery in the boot	photo
12:39:13	10:39:13	**MODES-RX-02 connected to the server** — first live data	server log
12:39:23	10:39	Battery display: input 0 W, output 9–10 W, 71 %, 13.8 h left	photo
12:39:39	10:39:39	Server health: *Feed RX-02 off → ok (Live — 24 messages/s)*	server log
12:39:49	10:39:49	Statistics page on the phone: MODES-RX-02 live	screenshot
12:40:05	10:40	*Feed RX-02: Live — 24 messages/s, connected 42 s . . . 22.2 % of its replies heard only here*	screenshot
12:40:49	10:40	Enclosure opened for the photos: receiver, Raspberry Pi 5, 4G router	photo
≈12:41:10	10:41	Power off — the last replies (260 in the last minute)	statistics
12:45:46	10:45	Everything packed in the boot	photo
12:46:21	10:46	Ready to drive	photo
15:03:54	13:03:54	Back on the balcony: **connected again by itself**, live at 68 messages/s	server log







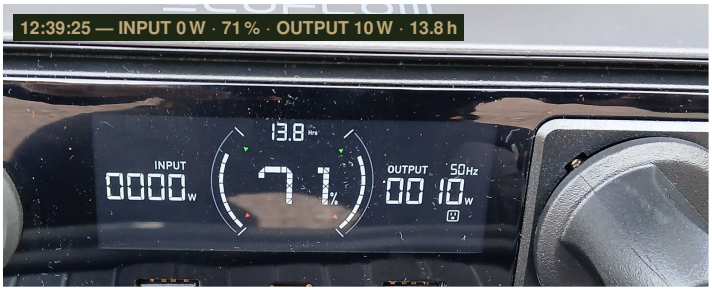


The station and its power



Inside the enclosure (left): the AirSpy receiver, the Raspberry Pi 5 and the mobile 4G router on one mounting plate, the coax coming in from the antenna on the mast above, and one power cable out. It is the same station that feeds as MODES-RX-02 from the balcony — nothing was changed for the test.

Power. An EcoFlow River 3 Plus (286 Wh, 230 V AC) stood in the open boot; a power cable ran across the gravel to the tripod. The battery's display a few seconds after the station had started:

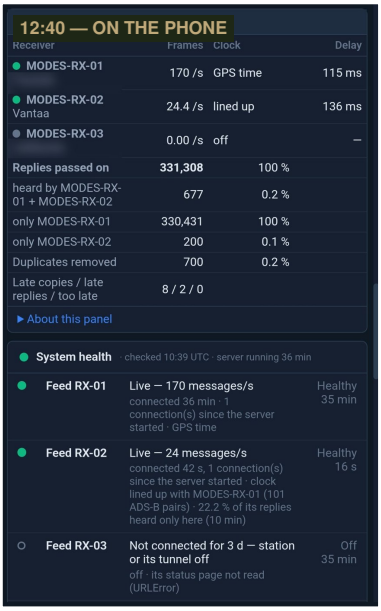


9–10 W for the whole station — receiver, computer and 4G router together, through the battery's inverter. At that load the battery's own estimate was 13.8 hours from 71 %, so roughly **a day from a full charge**; a car's 12 V socket or a small solar panel could keep it going longer (not yet tried).

What the station received

On the air (12:39:13 – ≈12:41:10)	≈2 min
Replies delivered to the server	2 648
ADS-B (DF 17): positions, velocities, identity	1 036
Comm-B (DF 20/21)	691
weather registers BDS 4,0 / 5,0 / 6,0	213 / 184 / 234
short replies (altitude, squawk, all-call, ACAS)	673
Aircraft	9
Highest aircraft	FL265
Rate (server health)	24 messages/s
Heard by no other receiver (server, 10 min)	22.2 %
On the balcony the same morning (07–11 local, hourly)	45–99 messages/s

The rate was lower than on the balcony: the antenna stood low in a forest clearing with trees around it, not high up on a building, and it was on the air for only two minutes on a Sunday noon. Even so, more than a fifth of what it heard reached the server through it alone — the point of a deployable receiver.



What the test shows — and what is next

- **Deployment is as quick as intended**: from the car to live data in under 8 minutes including the photos, and packed again in about 5.
- **Zero configuration really works away from home**: the station found the mobile network and the server by itself, at the site and again on the balcony.
- **Off-grid is easy**: about 10 W means a small portable battery runs it for a day.
- **Next**: a longer field day on a higher, open site (the reception range and the share of replies heard only there), the new Teltonika 4G router, and a direct 12 V supply instead of the 230 V inverter.

The test site is not named. The photos are the author's, edited for this report (location data removed; the car's number plate and receiver sites blurred). The figures are from the MODE-S Wind server's log and message statistics of 8 November 2026 and from the battery's display.